



Phase 2 Database Project

Hospital Management System

Team 4

Team Members

Name	ID
Saeed Bayoumy Nabawy	2200950
Zeyad Abdullah Shabaan	2200932
Aya Mounir Amer	2200758
Nadeen Ahmed El-Sayed	2200766
Lojain Mohamed Ibrahim	2201218
Hazem Hussein Hassan	2201337
Amen Hany Mohamed	2201404

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1. Project Theme

Hospital Management System

The project theme chosen for this application is a Hospital Management System. The purpose of the system is to help hospitals and clinics manage patient information, doctor records, appointments, and medical services in an organized and efficient way.

The application is connected to a relational database that stores all hospital-related data. The system allows users to perform important database operations such as inserting new records, viewing existing data, updating information, and deleting records when needed.

This project was developed to demonstrate the use of:

- Relational database design
- SQL queries and database interactions
- Windows Forms user interface
- CRUD operations (Create, Read, Update, Delete)
- Navigation between multiple forms

The system improves data organization inside the hospital environment.

2. Database Architecture

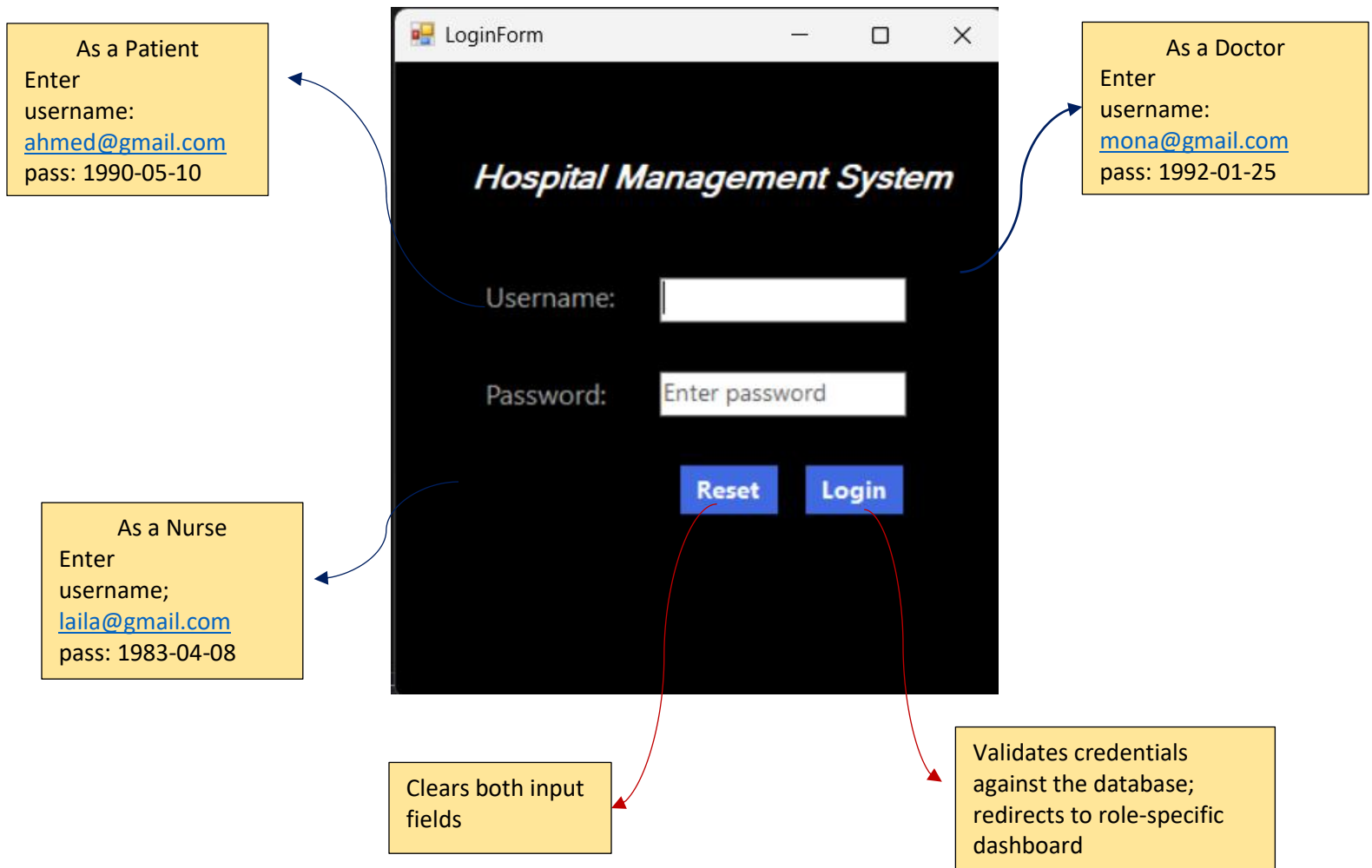
The database was designed using SQL Server and contains multiple related tables that represent the main entities inside the hospital system.

The database stores:

- Patient information
- Doctor information
- Nurse information
- Appointment details
- Medical records

3. Application Interface (UI)

1- Login Form

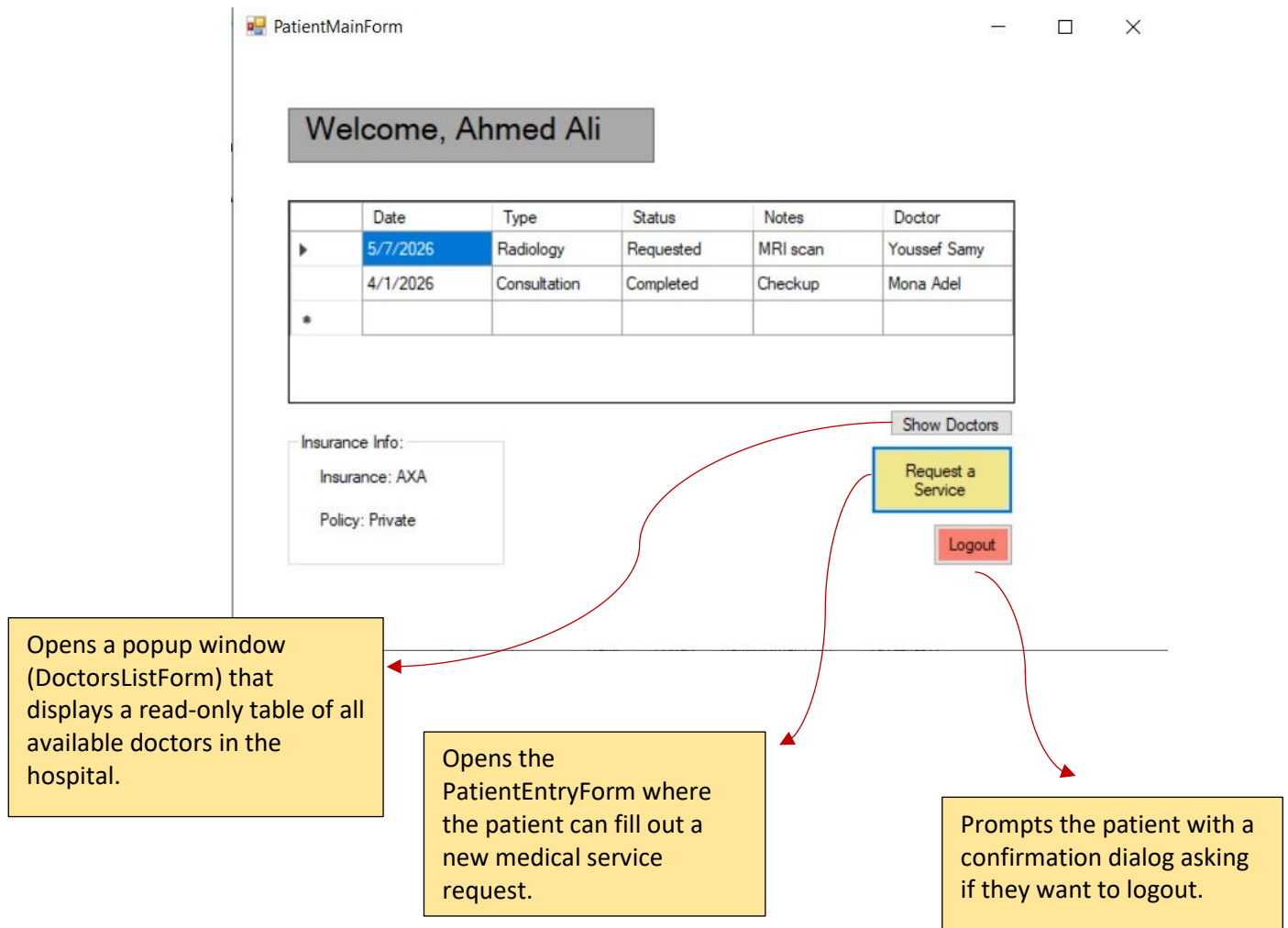


Functionality: Entry point.

Two fields: email and birth date (password).

Validates credentials and opens the correct dashboard based on role (Doctor, Patient, Nurse).

2- PatientMainForm



Functionality: The PatientMainForm is the patient dashboard.

1. After login, it shows the patient's name, email, and insurance details at the top.
2. The main area displays the patient's medical service history in a grid (date, type, status, notes, and assigned doctor). If no services exist, it shows "No medical history found" instead.

3- DoctorListForm (Read-Only)

	DoctorID	Name	Email	Shift	Department	Location
▶	4	Mona Adel	mona@gmail....	Morning	Cardiology	Floor 1
	5	Youssef Samy	youssef@gm...	Night	Neurology	Floor 2
	6	Nour Magdy	nour@gmail....	Morning	Emergency	Ground Floor

Close

This form displays a read-only list of all doctors registered in the system.

It shows a DataGridView with columns for doctor ID, full name, email, shift, department name, and department location.

The data is loaded directly from the database using a query that joins the Doctor, Person, and Department tables.

Dismisses the form and returns to the Patient Main Form

4- PatientEntryForm

The screenshot shows a web form titled "Request New Service" within a window labeled "PatientEntryForm". The form contains the following fields and controls:

- Service:** A dropdown menu with the text "-- Select Service Type --". A callout box explains: "A combo box that loads service types from the database. Patient picks the type of service they need."
- Preferred Doctor:** A dropdown menu with the text "-- Select a Doctor --". A callout box explains: "A combo box to select a doctor (loaded dynamically from the Doctor table)".
- Date:** A date picker showing "Thursday , May 7, 2026". A callout box explains: "A date picker for scheduling the appointment (cannot select past dates)".
- Notes:** A text input field. A callout box explains: "A text box for additional notes".
- Submit:** A green button. A callout box explains: "Button validates that a doctor and service type are selected, then inserts the service record into the database using a transaction to ensure data consistency. If any part fails, the transaction rolls back and an error message appears."
- Cancel:** A red button. A callout box explains: "Button closes the form, and it will ask for confirmation if any data was already entered."

Functionality: This form opens when a patient clicks "Request Service" on their dashboard. It allows patients to submit a new medical service request

5- DoctorMainForm

Doctor Main Form

Doctor Dashboard - Service Request

	ServiceID	ServiceDate	PatientID	Notes	Status
▶	1	4/1/2026	1	Checkup	Completed
	2	4/2/2026	2	MRI Scan	Completed
	3	4/3/2026	3	Emergency Case	Completed
	4	5/7/2026	1	MRI scan	Requested
*					

Assign Nurse: Service Type:

Changes the selected request status to "Accepted"

First, Dr. should select one of the services on dashboard
Then, select the wanted nurse's name

Only enabled when a request is already accepted; marks it as "Completed"

Inserts a new row into NurseServices and NurseHisServices to link the selected nurse and service type to the currently selected service request.

Button to reload the grid with the latest data from the database

button that returns to the login screen after confirmation

Updates the selected service status to "Rejected" in the database.

Any dr. selects wanted service type

Button that opens the patient management screen

Functionality: This is the main dashboard for doctors after they log in. The form displays a DataGridView showing all service requests from patients. Each row shows the service ID, date, patient ID, notes, and current status. The doctor can select any row and take action.

6- ManagePatientsForm

The screenshot shows a window titled "ManagePatientsForm" with a "Patient Management" header. It contains a DataGrid with the following data:

PersonID	Fname	Lname	Email	BirthDate	PhoneNumber
1	Ahmed	Ali	ahmed@gmail.com	5/10/1990	01000000001
2	Sara	Hassan	sara@gmail.com	3/15/1995	01000000002
3	omar	mohamed	omar@gmail.cok	5/7/2026	010292873
11	Lujain	Khalil	lojymohamed@g...	5/7/2026	0103357721

Below the grid is a form with fields for First Name, Last Name, Email, Phone Number, Insurance Company, Coverage, and Birthday. At the bottom are buttons: Add Patient, Delete Patient, Update Patient, Clear, Count, and Close.

Annotations:

- Removes the selected patient and all related records (services, insurance, phone) from the database after a confirmation prompt. (Points to Delete Patient button)
- Closes the form and returns to the DoctorMainForm. (Points to Close button)
- Creates a new patient record using the data entered in the text fields. Required fields are first name, last name, email, and phone; insurance fields are optional. (Points to Add Patient button)
- Displays a message box showing the total number of patients in the system. (Points to Count button)
- Modifies the selected patient's existing record after validating that required fields are filled. (Points to Update Patient button)
- Empties all text fields and deselects the current row. (Points to Clear button)

Functionality:

1. This form is accessed from the DoctorMainForm and allows doctors to manage the patient registry.
2. It shows a DataGridView listing all patients with columns for ID, first name, last name, email, birth date, phone number, and insurance details.
3. When a doctor clicks on any row, the patient's data fills the text boxes on the form for easy editing.

7- Nurse Main Form

The screenshot shows a window titled "Nurse Main Form" with a standard Windows title bar (minimize, maximize, close buttons). The main content area has a title "Assigned Services" in bold. Below the title is a table with the following data:

	ServiceDate	ServiceType	Notes	PatientName
▶	5/7/2026	Assistance	MRI scan	Ahmed Ali
	4/1/2026	Assistance	Checkup	Ahmed Ali

Below the table, there is a text label "Showing 2 service(s)." and two buttons: "Refresh" and "Logout".

Button reloads the task list from the database to show any newly assigned services.

Button closes the form and returns to the login screen.

Functionality: This is the dashboard for nurses.

It shows a DataGridView listing all services that have been assigned to the currently logged-in nurse.

Each row displays the service date, service type, notes, and patient name.

4. Navigation

1- Doctor wants to delete a patient

The screenshots illustrate the following steps:

- LoginForm**: The user enters the username `mona@gmail.com` and a password. A red arrow points to the **Login** button, with a label **Click here**.
- Doctor Main Form**: The **Doctor Dashboard - Service Request** table is shown. A red arrow points to the **Manage Patients** button, with a label **Click Here**.
- ManagePatientsForm**: The **Patient Management** table is shown. A red arrow points to the **Delete Patient** button, with a label **Click Here**.
- Confirm Delete**: A dialog box asks, "Are you sure you want to delete this patient and ALL related data?". A red arrow points to the **Yes** button.

The **Service Request** table data is as follows:

ServiceID	ServiceDate	PatientID	Notes	Status
1	4/1/2026	1	Checkup	Completed
2	4/2/2026	2	MRI Scan	Completed
3	4/3/2026	3	Emergency Case	Completed

The **Patient Management** table data is as follows:

PersonID	Fname	Lname	Email	BirthDate	PhoneNumber
1	Ahmed	Ali	ahmed@gmail.com	5/10/1990	01000000001
2	Sara	Hassan	sara@gmail.com	3/15/1995	01000000002
3	Omar	Khaled	omar@gmail.com	7/20/1988	01000000003

2- Patient wants to check available doctors or request a service

LoginForm

Hospital Management System

Username:

Password:

PatientMainForm

Welcome, Ahmed Ali

Date	Type	Status	Notes	Doctor
5/7/2026	Radiology	Requested	MRI scan	Youssef Samy
4/1/2026	Consultation	Completed	Checkup	Mona Adel

Insurance Info:
Insurance: AXA
Policy: Private

Available Doctors

DoctorID	Name	Email	Shift	Department	Location
4	Mona Adel	mona@gmail...	Morning	Cardiology	Floor 1
5	Youssef Samy	youssef@gm...	Night	Neurology	Floor 2
6	Nour Magdy	nour@gmail...	Morning	Emergency	Ground Floor

PatientEntryForm

Request New Service

Service:

Preferred Doctor:

Date:

Notes: